It sounds like you're looking for **courses/resources that teach iterative hardware prototyping with early evaluation, PCB layout, and Altium Designer workflows**. That's a great focus—this is exactly how modern hardware teams move fast without costly redesigns.

Let me break this into **(1) what to learn**, **(2) good courses**, and **(3) a practical learning path**.

🧠 1. What "iterative hardware prototyping" actually involves

To do this well, you need to combine:

🔁 Iterative design mindset

- Build → Test → Improve → Repeat
- Early validation (before full PCB)
- Use dev boards, breadboards, and simulation first

⚡ Early evaluation techniques

- Breadboarding / perfboard testing
- Using eval kits (MCU, sensors, PMICs)
- Simulation tools (SPICE)
- Signal integrity checks early

🧩 PCB layout skills

- Component placement strategy
- Routing (power vs signal)
- Ground planes, decoupling
- EMI/EMC basics

🛠️ Tools (key one you mentioned)

- Altium Designer
- LTspice
- KiCad (good beginner alt)

🎓 2. Best courses (focused on Altium +

practical PCB + prototyping)

🟢 Beginner → Intermediate (strong foundation)

1. Udemy – Altium + PCB design

- "Altium Designer PCB Design Course" (various instructors)
- Covers:
  - Schematic → PCB → fabrication
  - Real projects
- Good for: tool mastery

---

2. Altium Academy

- Best structured learning for Altium
- Topics:
  - Layout best practices
  - High-speed design basics
  - Component libraries
- More professional-grade than Udemy

---

3. Phil's Lab

- One of the **best real-world PCB designers online**
- Teaches:
  - Iterative design approach
  - Debugging boards
  - RF + mixed signal layout
- Very practical, industry-style thinking

🔵 Intermediate → Advanced (iteration + real engineering)

4. Contextual Electronics

4. Contextual Electronics

- Focus: **real product development workflow**
- Covers:
  - Prototyping cycles
  - Design reviews
  - Testing methodology
- Closest to actual startup/industry flow

---

5. MIT OpenCourseWare

Look for:

- Electronics / embedded systems courses
- Teaches:
  - Design thinking
  - System-level validation

---

6. YouTube deep dives

- EEVblog
- Robert Feranec

Focus on:

- PCB reviews
- Mistakes & redesigns (very valuable for iteration learning)

🧭 3. Suggested learning roadmap (practical + iterative)

Phase 1: Basics (2–3 weeks)

- Learn:
  - Schematics
  - Basic PCB layout

    - Basic PCB layout

- Tool:
    - Altium Designer
- Project:
    - LED driver board
    - Simple power supply

Phase 2: First iteration loop (key step 🚀)

- Build:
    - Sensor + microcontroller board
- Do:
    - Prototype on breadboard first
    - Then design PCB
- Evaluate:
    - Power noise
    - Signal errors
- Revise PCB → Version 2

Phase 3: Advanced layout thinking

- Learn:
    - Ground planes
    - Decoupling strategy
    - High-speed traces
- Project:
    - USB / communication board

Phase 4: Real-world workflow

- Introduce:

- Introduce:
  - Design reviews
  - Checklists
  - Version control (Git for hardware)
- Try:
  - 2–3 board iterations intentionally

🔁 Key mindset shift (most important)

Most beginners:

> "Design once → hope it works"

Good hardware engineers:

> "Design quickly → test early → expect failure → iterate fast"

💡 Pro tip (industry trick)

Before designing PCB:

- Always validate:
  - Power circuit
  - Critical ICs
- Use eval boards + jumper wires → Saves weeks of debugging later

---

If you want, I can:

- Suggest a **complete hands-on project roadmap (with parts list)**
- Or recommend **India-friendly component suppliers + PCB fabs**
- Or build a **90-day learning plan tailored to you**